

Julius Alexander Zeiss, M.Sc. Physics

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Profile

- I develop classical algorithms that make hard quantum problems tractable. Drawing on group theory, quantum Shannon theory, and convex optimization, I design methods with provable guarantees for studying quantum correlations and error correction. My work uses quantum de Finetti theorems and symmetry-reduced semidefinite programming hierarchies to obtain rigorous bounds for problems that are otherwise intractable or, in some settings, even undecidable. I complement these approximation methods with exact results on the limits of entanglement-assisted communication.
- Speaker at leading international conferences (TQC, Beyond IID, AQIS) and invited speaker at institutions including Stanford University and the Perimeter Institute for Theoretical Physics
- Host of a regular A.I. Workshop at RWTH Aachen on AI tools for rigorous, reproducible science

Education

since 03/2023	PhD in Physics (Quantum Information Theory), RWTH Aachen University, Germany Institute for Quantum Information, final-year PhD student Advisor: Prof. Dr. Mario Berta
10/2019 – 08/2022	M.Sc. Physics, University of Cologne & University of Bonn, Germany Focus: Theoretical Particle Physics, General Relativity, Mathematical Physics Thesis: <i>SDP Relaxations for Quantum Inflation</i> Advisor: Prof. Dr. David Gross, Thesis Grade: 1.0 (very good) Overall Grade: 1.4 (very good)
10/2018 – 08/2019	M.Sc. Physics, University of Münster, Germany Focus: Theoretical Nuclear & Particle Physics, Theoretical Non-linear Physics, Quantum Information Theory, Biophysics
10/2015 – 09/2018	B.Sc. Physics, University of Münster, Germany Thesis: <i>Numerical Treatment of the Manning Potential for the Description of Ammonia</i> Advisor: Prof. Dr. Gernot Münster, Thesis Grade: 1.0 (very good) Overall Grade: 1.8 (good)
04/2013 – 09/2015	B.Sc. Business Administration, University of Münster, Germany (discontinued, transferred to Physics)
09/2010 – 05/2012	International Baccalaureate, King William's College, Isle of Man Overall Grade: 41/45 Points (very good, 1.1 German Abitur)

Publications and Preprints

2026	J. A. Zeiss, <i>Optimal entanglement-assisted source coding under a balanced-difference promise</i> , arXiv:2609.15757
2026	M. Berta, P. Costa Rico, G. Koßmann, L. Lami, J. A. Zeiss, <i>Sharp continuity of quantum conditional entropy</i> , arXiv:2607.24687
2026	G. Koßmann, J. A. Zeiss, <i>Fixed points in de Finetti hierarchies</i> , arXiv:2607.23689 (under review at Communications in Mathematical Physics)
2026	J. A. Zeiss, G. Koßmann, R. Schwonnek, M. Plávala, <i>Finite de Finetti for convex bodies and polynomial optimization</i> , arXiv:2601.15184 (under review at Mathematical Programming A)
2025	J. A. Zeiss, G. Koßmann, O. Fawzi, M. Berta, <i>Approximating fixed size quantum correlations in polynomial time</i> , arXiv:2507.12302 (under review at PRX Quantum)
2025	G. Koßmann, J. A. Zeiss, O. Fawzi, M. Berta, <i>On approximate quantum error correction for symmetric noise</i> , arXiv:2507.12326 (under review at Quantum)

Invited Talks

06/2026	Stanford University, CS Theory Group, Palo Alto, USA <i>A Family of Information-theoretic de Finetti Theorems for Constrained Optimization</i>
04/2026	University of Cologne, Quantum Information Seminar, Germany <i>Approximating fixed size quantum correlations in polynomial time</i>
07/2023	Siegen University, Germany <i>Inner approximation schemes for approximate quantum error correction</i>
02/2023	Perimeter Institute for Theoretical Physics, Waterloo, Canada <i>Infinite Dimensional Optimisation Problems in Quantum Information — An operator algebra approach to the NPA Hierarchy</i>
05/2022	Alpen-Adria-Universität Klagenfurt, Austria <i>SDP methods for the quantum causal compatibility problem</i>

Contributed Talks

09/2026	TQC 2026, Sherbrooke, Canada <i>A Family of Information-Theoretic de Finetti Theorems for Constrained Optimization</i>
05/2026	ILAS 2026, Virginia Tech, Blacksburg, USA <i>Convergent Inner-Outer Approximation Schemes From De Finetti Theorems For Games And Quantum Error Correction</i>
03/2026	Quantum at the Dunes, IIP-UFRN, Natal, Brazil <i>Approximating fixed size quantum correlations in polynomial time</i>
08/2025	AQIS 2025, The University of Hong Kong, Hong Kong <i>Approximating fixed size quantum correlations in polynomial time</i>
07/2025	13th Beyond IID in Information Theory, TU Munich, Germany <i>Approximating fixed size quantum correlations in polynomial time</i>

Posters

04/2026	851. WE-Heraeus-Seminar: 90 Years of Einstein-Podolsky-Rosen Paradox, Bad Honnef, Germany <i>Approximating fixed size quantum correlations in polynomial time</i>
01/2026	QIP 2026, Riga, Latvia <i>Approximating fixed size quantum correlations in polynomial time</i>
08/2025	QMATH Masterclass on Representation Theory in Quantum Information Science, University of Copenhagen, Denmark <i>Schur–Weyl Duality for Efficient Approximation of Constrained Separability</i>
07/2024	Beyond IID in Information Theory, University of Illinois Urbana-Champaign, USA <i>Approximating fixed size quantum correlations in polynomial time</i>

Teaching

since 03/2023	RWTH Aachen University, Germany Teaching and supervision of students at the Institute for Quantum Information
04/2021 – 04/2022	University of Cologne, Germany Tutorials on Linear Algebra & Analysis, Proofs in Mathematics, Theoretical Physics
04/2019 – 09/2019	University of Münster, Germany Tutorials on Applied Physics

Academic Service

Conferences	TQC (2023, 2024, 2025) • ISIT (2023, 2024, 2025) • QIP (2024) • Beyond IID (2023, 2024)
Journals	IEEE (2025)

Outreach

since 2025	A.I. Workshop, RWTH Aachen University, Germany Founder and host of a regular workshop on AI tools for research — from coding assistants and version control to formally verified proofs — and on using them to conduct better science
2024 – 2026	ML4Q — Matter and Light for Quantum Computing, Germany Instructor: school internships (July 2024), Quanteen Day 2024, Quanteen Day 2026
11/2022	DPG Physik Journal A. Matthies and J. Zeiss, <i>Quantum Computing</i> , review article on the Quantum Computing Bad Honnef Physics School, Physik Journal 21 (2022) No. 11, p. 57

Skills

Languages	German (native), English (bilingual)
Coding	Python, Mathematica, Fortran, PHP, JavaScript, HTML, SQL, VBA

Outside of Academia — Employment

2014, 2015	Fielmann AG, Hamburg, Germany — DCF Modelling with Monte-Carlo Simulations, VBA & SQL Solutions
01/2013 – 02/2013	HMT, Hamburg, Germany — Mathematical Modelling using Monte-Carlo Simulations
08/2011	AKO Capital, London, England — Internship in Financial Research using Mathematical Modelling